1. Let $\mathbf{F}(x,y) = \langle P(x,y), Q(x,y)\rangle$ be a conservative vector field on a connected domain D. For each statement below, write **T** (true) or **F** (false). **(3 points each)**

(a) ☐ $\int_C \mathbf{F}\cdot d\mathbf{r}$ depends only on the endpoints of C.

(b) ☐ $\oint_C \mathbf{F}\cdot d\mathbf{r} = 0$ for every closed curve C in D.

(c) ☐ $\dfrac{\partial P}{\partial x} = \dfrac{\partial Q}{\partial y}$ at every point in D.

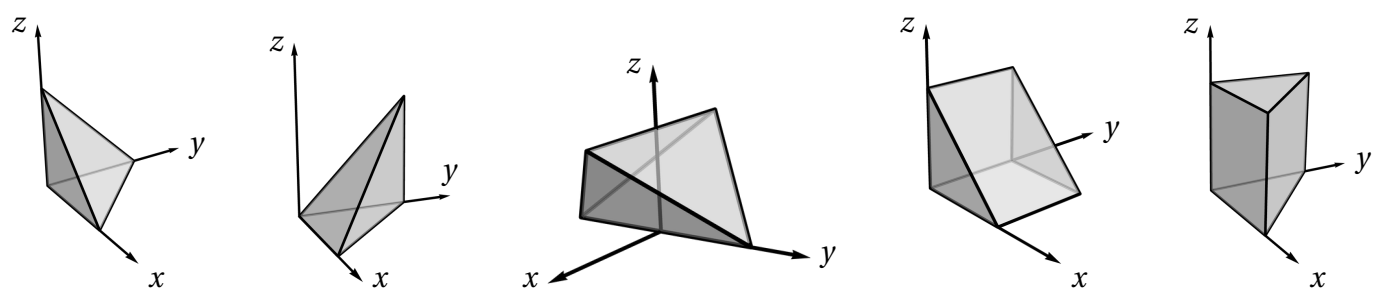
(d) ☐ There exists a unique function f such that $\mathbf{F} = \nabla f$.

(e) ☐ $\int_C P(x,y)\,dx = 0$ for every closed curve C in D.

2. Consider the following integral:

$$\int_0^1 \int_0^{1-x} \int_0^{1-x-y} f(x,y,z)\,dz\,dy\,dx.$$

(i) Circle the figure that best represents the region of integration. **(7 points)**



(ii) Circle the integral that represents the same region using the order $dx\,dz\,dy$. **(8 points)**

$$\int_0^1 \int_0^{1-z} \int_0^{1-y-z} f(x,y,z)\,dx\,dz\,dy \qquad \int_0^1 \int_0^{1-y} \int_0^{1-y-z} f(x,y,z)\,dx\,dz\,dy$$

$$\int_0^1 \int_0^{1-y} \int_0^{1-x-y} f(x,y,z)\,dx\,dz\,dy \qquad \int_0^1 \int_0^{1-z} \int_0^{1-x-z} f(x,y,z)\,dx\,dz\,dy$$

Scratch Space

3. Consider the vector field

$$\mathbf{F}(x, y, z) = \langle y + z,\ x - z,\ x + y^2 \rangle$$

on $\mathbb{R}^3$.

(a) Circle the curl of **F**: **(5 points)**

$\langle 1, -1, 1 \rangle$	$\langle 1, 0, 1-2y \rangle$	$\langle 1, -1, 2y-1 \rangle$	$\langle 0, -1, 2y \rangle$	$\langle 2y+1, 0, 0 \rangle$

(b) Circle the divergence of **F**: **(5 points)**

$2y$	$1+2y$	0	2	$x+y+z$

4. Consider the surface

$$z = x^2 + y^2$$

over the region $D = \{(x, y) : x^2 + y^2 \leq 1\}$.

What is the surface area of S? **(5 points)**

$\int_0^{2\pi} \int_0^1 \sqrt{1+r^2}\, r\, dr\, d\theta$	$\int_0^{2\pi} \int_0^1 \sqrt{1+4r^2}\, r\, dr\, d\theta$	$\int_0^{2\pi} \int_0^1 \sqrt{1+r^2}\, dr\, d\theta$

5. Let C denote the triangle with vertices $(0,0)$, $(2\pi,0)$, and $(0,2\pi)$, oriented **anticlockwise**.

Use Green's theorem to compute

$$\int_C \left(e^y \sin(x) + 2y\cos(x)\right) dx + \left(2\sin(x) - e^y \cos(x) + 2x\right) dy.$$

(10 points)

$$\int_C \left(e^y \sin(x) + 2y\cos(x)\right) dx + \left(2\sin(x) - e^y \cos(x) + 2x\right) dy =$$

Scratch Space

6. Consider the region R which is bounded below by the cone $z = \sqrt{x^2 + y^2}$ and which is bounded above by the sphere $x^2 + y^2 + z^2 = 1$.

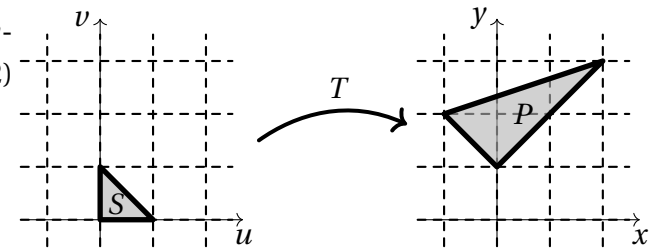
(a) **Using Cylinderical coordinates**, fill in the limits and integrand of the iterated integral below to compute the volume of R. Use the order of integration $dz\,dr\,d\theta$. **(8 points)**

$$\text{Volume of } R = \int \qquad \int \qquad \int \qquad dz\,dr\,d\theta$$

(b) **Using spherical coordinates,** fill in limits and integrand of the triple integral below so that it computes the volume of R. Use the order of integration $d\rho\,d\phi\,d\theta$. **(8 points)**

$$\text{Volume of } R = \int \qquad \int \qquad \int \qquad d\rho\,d\phi\,d\theta$$

7. Consider the triangles S and P shown on the right. Triangle S has vertices $(0,0)$, $(1,0)$, $(0,1)$ in the uv-plane. Triangle P has vertices $(0,1)$, $(2,3)$, and $(-1,2)$ in the xy-plane.



(i) Circle the transformation $T(u,v)$ that maps triangle S onto triangle P. **(5 points)**

$T(u,v) = (u-v,\ 1+u+v)$ $\qquad$ $T(u,v) = (2u-v,\ 1+2u+v)$

$T(u,v) = (-u+2v,\ 1+u+2v)$ $\qquad$ $T(u,v) = (u+v,\ 1+2u+2v)$

(ii) Using the transformation from part (i), set up (but do not evaluate) an integral that computes

$$\iint_P y\cos(x)\,dA$$

as an integral over S. **(10 points)**

$$\boxed{\iint_P y\cos(x)\,dA = \int \qquad \int \qquad\qquad\qquad du\,dv}$$

8. Consider $\mathbf{F}(x, y, z) = (\sin(yz),\ xz\cos(yz),\ xy\cos(yz) + z)$.

(i) Find a potential function f so that $\nabla f(x, y, z) = \mathbf{F}(x, y, z)$ on $\mathbb{R}^3$. **(10 points)**

$\nabla f = \mathbf{F}$ for $f(x, y, z) =$

(ii) Curve C has parametrization $\mathbf{r}(t) = (t, 1 - t, 2t)$ for $t \in [0, 1]$. Compute $\int_C \mathbf{F} \cdot d\mathbf{r}$. **(5 points)**

$$\int_C \mathbf{F} \cdot d\mathbf{r} =$$